

Lecture 6

USING SIMULATION TO EVALUATE
AND EXTEND OUR METHODS

ADF test critical values

ADF test critical values: **type="drift"**

$$\Delta Y_t = \beta_0^* + \varphi Y_{t-1} + U_t$$

$$H_0 : \varphi = 0 \quad \text{vs} \quad H_1 : \varphi < 0$$

```
1 library(urca)
2 SignificanceLevels <- c(0.01, 0.05, 0.1)
3 cv_drift <- qunitroot(p=SignificanceLevels,
4                       trend="c", statistic="t")
```

	1%	5%	10%
cv_drift	-3.430	-2.861	-2.567

ADF test critical values: **type="trend"**

$$\Delta Y_t = \beta_0^* + \beta_1^* t + \varphi Y_{t-1} + U_t$$

$$H_0 : \varphi = 0 \quad \text{vs} \quad H_1 : \varphi < 0$$

```
1 cv_trend <- qunitroot(p=SignificanceLevels,  
2                       trend="ct", statistic="t")
```

	1%	5%	10%
cv_trend	-3.958	-3.410	-3.127

ADF test critical values

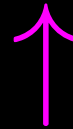
	1%	5%	10%
cv_drift	-3.430	-2.861	-2.567
cv_trend	-3.958	-3.410	-3.127

- These are *asymptotic* (approximate) critical values. Like “1.96” for standard t tests.
- Approximate critical values for any n can also be found. Like t critical values.

ADF test critical values

Example. $n = 100$

```
1 cv_drift <- qunitroot(p = SignificanceLevels,  
2                      trend = "c", N = 100, statistic = "t")
```



	1%	5%	10%
cv_drift	-3.497	-2.891	-2.582

ADF test critical values

cv_drift

	1%	5%	10%
n=50	-3.568	-2.921	-2.599
n=100	-3.497	-2.891	-2.582
n=200	-3.463	-2.876	-2.574
n=400	-3.447	-2.869	-2.571
n=800	-3.438	-2.865	-2.569
n=Infinity	-3.430	-2.861	-2.567

Practically, allowing critical values to vary with n makes very little difference.

ADF test critical values

cv_trend

	1%	5%	10%
n=50	-4.153	-3.502	-3.181
n=100	-4.052	-3.455	-3.153
n=200	-4.005	-3.432	-3.140
n=400	-3.981	-3.421	-3.133
n=800	-3.969	-3.415	-3.130
n=Infinity	-3.958	-3.410	-3.127

Practically, allowing critical values to vary with n makes very little difference.

ADF test critical values

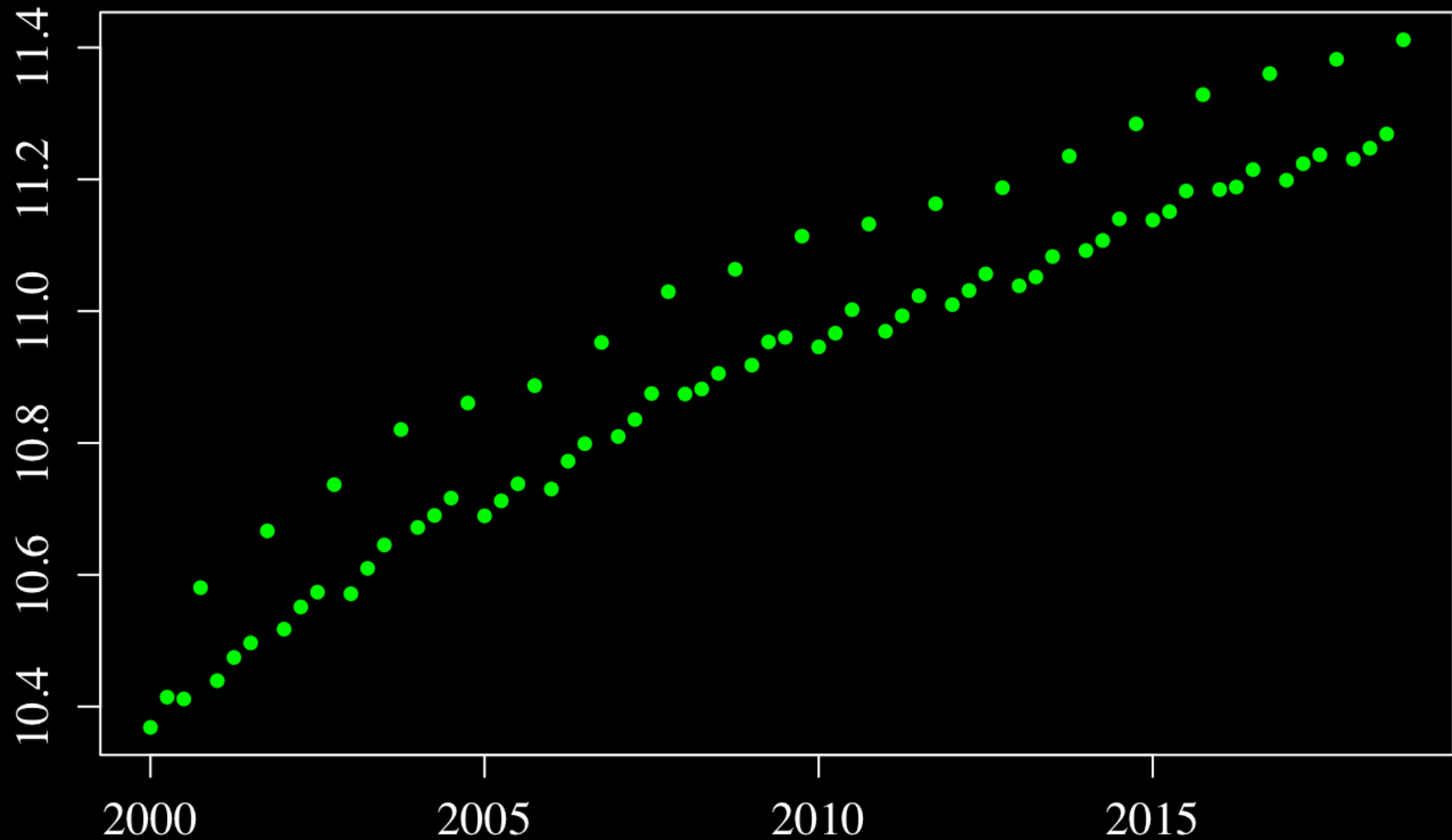
	1%	5%	10%
cv_drift	-3.430	-2.861	-2.567
cv_trend	-3.958	-3.410	-3.127

- *Different* critical values for different trend functions.

This is not the case for standard tests.

- These differences are large enough to matter in practice. We need the right critical value for the trend function used!

log of Retail Sales, 2000q1 - 2018q4



Trend function for $Y_t = \log \text{Retail Sales}$:

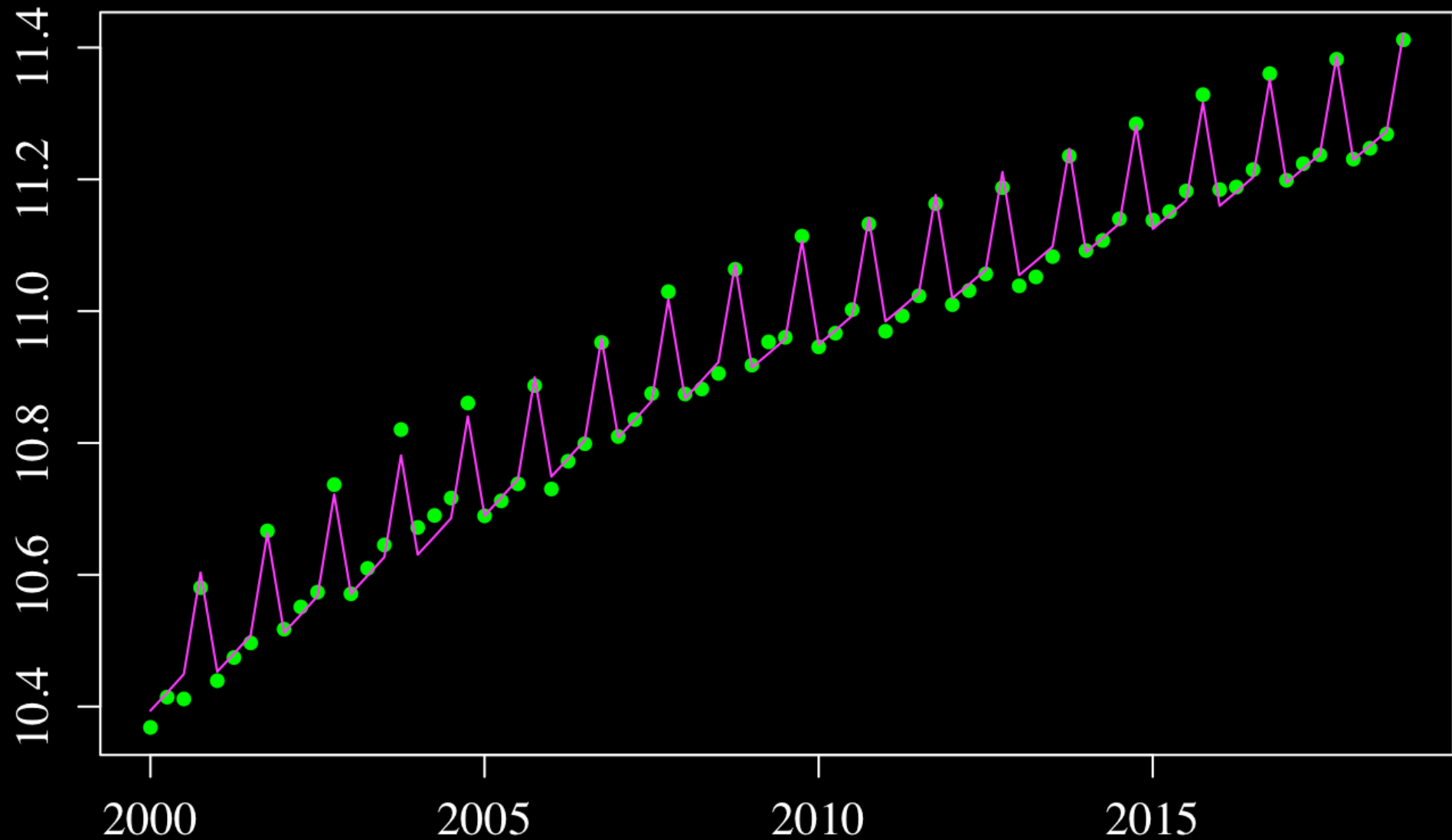
$$Y_t = \beta_0 + \beta_1 \text{Time}_t + \beta_2 \text{TimePostGFC}_t \\ + \delta_1 Q_{1,t} + \delta_2 Q_{2,t} + \delta_3 Q_{3,t} + Z_t$$

where

$$\text{TimePostGFC}_t \\ = (\text{Time}_t - 2008.5) \times 1(\text{Time}_t > 2008.5)$$

How to test for a unit root in Z_t ?

log of Retail Sales, 2000q1 - 2018q4



Deviations from trend function: $\hat{Z}_t$



Trend function for $Y_t = \log \text{Retail Sales}$:

$$Y_t = \beta_0 + \beta_1 \text{Time}_t + \beta_2 \text{TimePostGFC}_t \\ + \delta_1 Q_{1,t} + \delta_2 Q_{2,t} + \delta_3 Q_{3,t} + Z_t$$

- **ur.df** and **qunitroot** do not allow for custom trend functions

Trend function for $Y_t = \log \text{Retail Sales}$:

$$Y_t = \beta_0 + \beta_1 \text{Time}_t + \beta_2 \text{TimePostGFC}_t \\ + \delta_1 Q_{1,t} + \delta_2 Q_{2,t} + \delta_3 Q_{3,t} + Z_t$$

- **ur.df** and **qunitroot** do not allow for custom trend functions
- We need to **calculate the ADF test** allowing for this trend function.
- We need to find the **appropriate critical value** for this trend function.

What is a critical value anyway?

Components of a generic hypothesis test

- Hypotheses: H_0 vs H_1
- Significance level α
- A test statistic T_n
- A decision rule such as “reject H_0 if $T_n < c_\alpha$ ”

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- Hypotheses: H_0 vs H_1
- Significance level α
- A test statistic T_n
- A decision rule such as “reject H_0 if $T_n < c_\alpha$ ”
(or “reject H_0 if $T_n > c_\alpha$ ”
or “reject H_0 if $|T_n| > c_\alpha$ ”)

Components of a generic hypothesis test

- A decision rule such as “reject H_0 if $T_n < c_\alpha$ ”

Critical value with significance level α :

$$P(T_n < c_\alpha \mid H_0 \text{ true}) = \alpha$$

Components of a generic hypothesis test

- A decision rule such as “reject H_0 if $T_n < c_\alpha$ ”

Critical value with significance level α :

$$P(T_n < c_\alpha \mid H_0 \text{ true}) = \alpha$$

- Reject H_0 when H_0 is true = Type 1 Error

Components of a generic hypothesis test

- A decision rule such as “reject H_0 if $T_n < c_\alpha$ ”

Critical value with significance level α :

$$P(T_n < c_\alpha \mid H_0 \text{ true}) = \alpha$$

- **Frequentist probability**: proportion of samples in which H_0 is rejected in repeated draws from the population distribution when H_0 is true.
- **We can simulate this probability!**

Example: simulation of ADF critical value

Simulated draw from a random walk

```
1 Y <- arima.sim(n=100,  
2       model=list(order=c(0,1,0)))
```

Time Series:

Start = 0

End = 100

Frequency = 1

[1]	0.000	1.371	0.806	1.169	1.802
[6]	2.207	2.100	3.612	3.517	5.536
[11]	5.473	6.778	9.064	7.676	7.397
[16]	7.264	7.899	7.615	4.959	2.518
[21]	3.838	3.532	1.750	1.579	2.793
[26]	4.688	4.258	4.001	2.238	2.698
[31]	2.058	2.513	3.218	4.253	3.644
[36]	4.149	2.432	1.648	0.797	-1.618

Simulated draw from a random walk

```
1 Y <- arima.sim(n=100,  
2      model=list(order=c(0,1,0)))
```

Time Series:

Start = 0

End = 100

Frequency = 1

↑
ARIMA(0, 1, 0)

[1]	0.000	1.371	0.806	1.169	1.802
[6]	2.207	2.100	3.612	3.517	5.536
[11]	5.473	6.778	9.064	7.676	7.397
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```

Time Series:

Start = 0

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Sample size

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[6]	2.207	2.100	3.612	3.517	5.536
[11]	5.473	6.778	9.064	7.676	7.397
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Simulated draw from a random walk

```
1 Y <- arima.sim(n=100,  
2     model=list(order=c(0,1,0)))
```

Time Series:

Start = 0 ← includes $Y_0 = 0$

End = 100

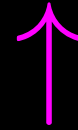
Frequency = 1



[1]	0.000	1.371	0.806	1.169	1.802
[6]	2.207	2.100	3.612	3.517	5.536
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[36]	4.149	2.432	1.648	0.797	-1.618

Simulated draw from a random walk

```
1 Y <- arima.sim(n=100,  
2   model=list(order=c(0,1,0)))[-1]
```



Remove Y_0

Simulated draw from a random walk

```
1 Y <- arima.sim(n=100,  
2      model=list(order=c(0,1,0)))[-1]
```

```
[1] 1.371 0.806 1.169 1.802 2.207  
[6] 2.100 3.612 3.517 5.536 5.473  
[11] 6.778 9.064 7.676 7.397 7.264  
[16] 7.899 7.615 4.959 2.518 3.838  
[21] 3.532 1.750 1.579 2.793 4.688  
[26] 4.258 4.001 2.238 2.698 2.058  
[31] 2.513 3.218 4.253 3.644 4.149  
[36] 2.432 1.648 0.797 -1.618 -1.581  
[41] -1.375 -1.736 -0.978 -1.705 -3.073  
[46] -2.640 -3.452 -2.008 -2.439 -1.784  
[51] -1.462 -2.246 -0.670 -0.027 0.063  
[56] 0.339 1.019 1.109 -1.885 -1.600
```

Simulated draw from a random walk

```
1 Y <- arima.sim(n=100,  
2               model=list(order=c(0,1,0)))[-1]
```

$$\text{ARIMA}(0, 1, 0) : \quad Y_t = Y_{t-1} + U_t$$

Satisfies $H_0 : \phi_1 = 1$ (unit root) in

$$Y_t = \phi_1 Y_{t-1} + U_t$$

ADF test on simulated random walk

```
1 Y <- arima.sim(n=100,  
2       model=list(order=c(0,1,0)))[-1]  
3 ADF <- ur.df(Y, type="drift", lags=0)
```

Test equation:

	Estimate	Std. Error	t value	Pr(> t)
drift	0.1755	0.1326	1.3236	0.1887
z.lag.1	-0.0701	0.0374	-1.8745	0.0639



Test statistic : $T_n = -1.8745$

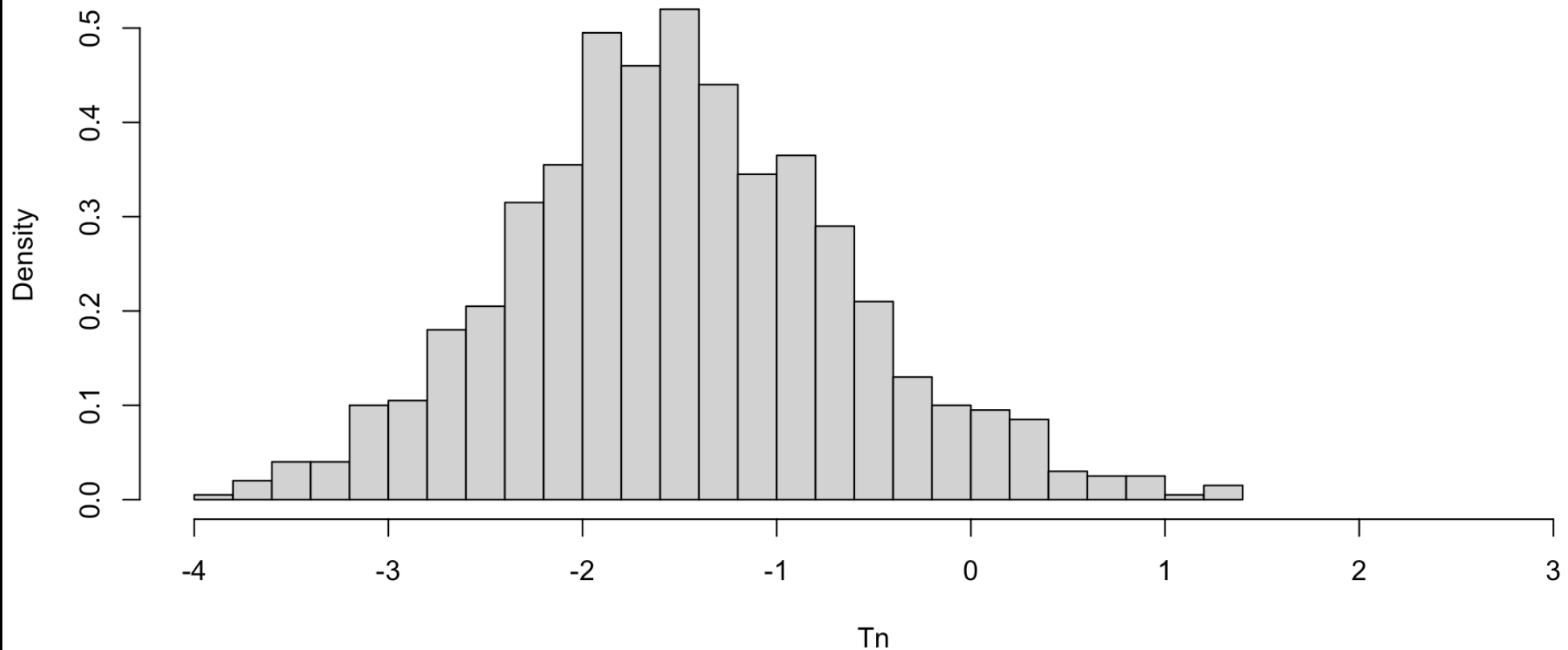
Repeat ADF tests on simulated random walks

```
1 reps <- 1000
2 Tn <- matrix(nrow=reps, ncol=1)
3
4 # Repeat reps times
5 for (r in 1:reps){
6
7   # Simulate random walk
8   Y <- arima.sim(n=100,
9     model=list(order=c(0,1,0)))[-1]
10
11   # Carry out ADF test
12   ADF <- ur.df(Y, type="drift", lags=0)
13   Tn[r] <- ADF@teststat[1]
14 }
```

Sampling distribution of ADF test

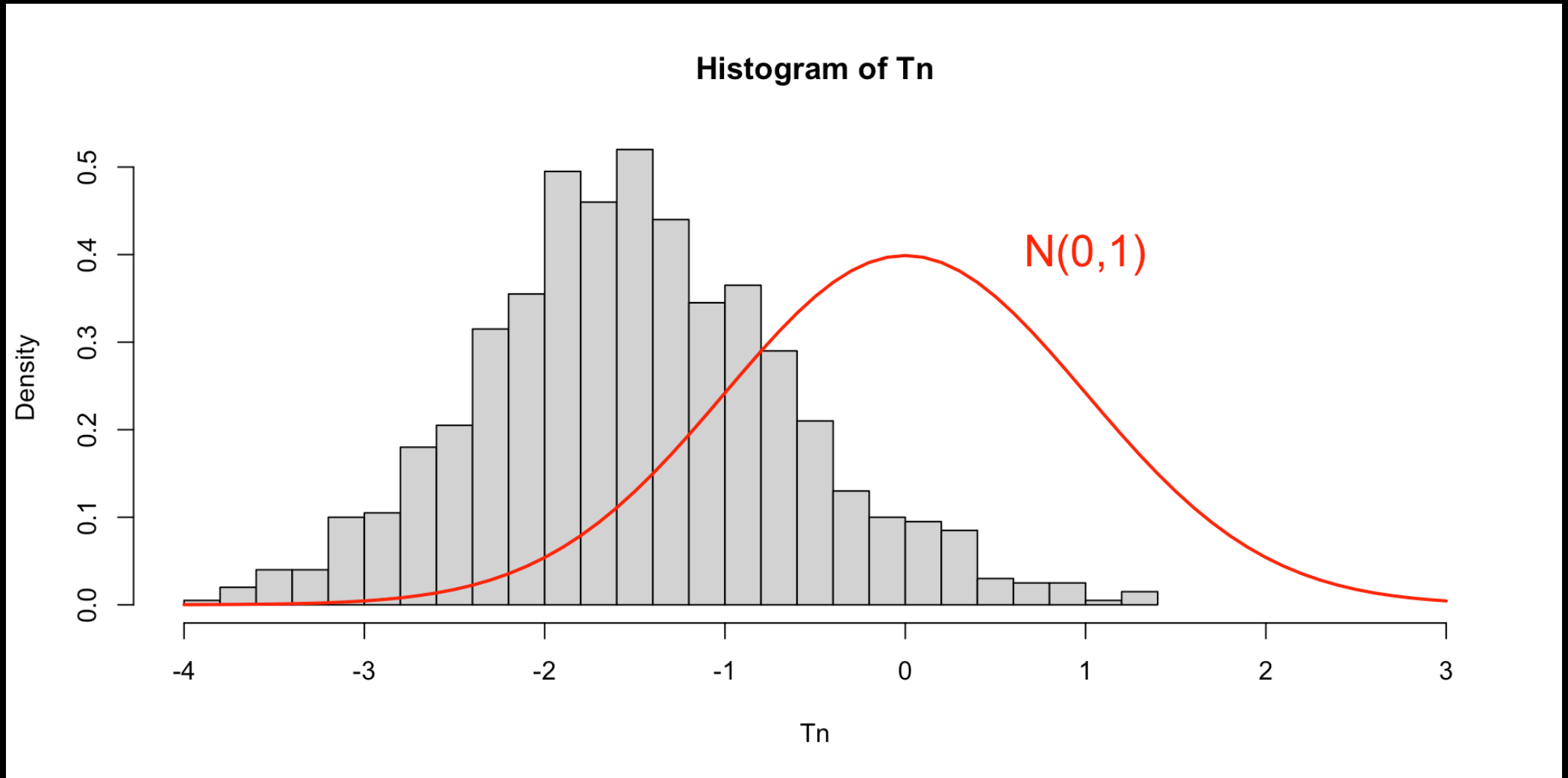
```
1 hist(Tn)
```

Histogram of Tn



Sampling distribution of ADF test

```
1 hist(Tn)
```



Simulated critical values of ADF test

```
1 SignificanceLevels <- c(0.01, 0.05, 0.1)
2 cv_sim <- quantile(Tn,
3                     probs=SignificanceLevels)
```

	1%	5%	10%
cv_sim	-3.462	-2.894	-2.587

Comparison:

```
1 cv_drift <- qunitroot(p=SignificanceLevels,
2                       N=100, trend="c", statistic="t")
```

	1%	5%	10%
cv_drift	-3.497	-2.891	-2.582

What is a p-value anyway?

Components of a generic hypothesis test

- Hypotheses: H_0 vs H_1
- Significance level α
- A test statistic T_n
- A decision rule such as “reject H_0 if $T_n < c_\alpha$ ”

Suppose we calculate the test statistic to be t .

$$p\text{-value : } p = P(T_n < t \mid H_0 \text{ true})$$

Decision rule: reject H_0 if $p < \alpha$.

Simulated p -value of ADF test

Eg. suppose we calculate $t = -1.8$.

Then $p = P(T_n < -1.8 \mid H_0 \text{ true}) :$

```
1 p_sim = mean(Tn < -1.8)
```

The proportion of the simulated statistics under H_0 that satisfy $T_n < -1.8$.

```
p_sim = 0.372
```

Simulated p -value of ADF test

Eg. suppose we calculate $t = -1.8$.

Then $p = P(T_n < -1.8 \mid H_0 \text{ true}) :$

```
1 p_sim = mean(Tn < -1.8)
```

```
p_sim = 0.372
```

Comparison:

```
1 p = punitroot(-1.8, N=100, trend="c",  
2               statistic="t")
```

```
p      = 0.379
```

Notes on the ADF simulation process

$$\Delta Y_t = X_t' \beta^* + \varphi Y_{t-1} + \sum_{j=1}^{p-1} \psi_j \Delta Y_{t-j} + U_t$$

- The specification of X_t matters most.
- n has minor effect.
- p and $\psi_1, \dots, \psi_{p-1}$: “asymptotically negligible”.
- Distribution of U_t : “asymptotically negligible”.

Notes on the ADF simulation process

Therefore under H_0 we simulate from

$$\Delta Y_t = X_t' \beta^* + \varphi Y_{t-1} + \sum_{j=1}^{p-1} \psi_j \Delta Y_{t-j} + U_t$$

Notes on the ADF simulation process

Therefore under $H_0 : \varphi = 0$ we simulate from

$$\Delta Y_t = X_t' \beta^* + \varphi Y_{t-1} + \sum_{j=1}^{p-1} \psi_j \Delta Y_{t-j} + U_t$$

Notes on the ADF simulation process

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$$\Delta Y_t = X_t' \beta^* + \sum_{j=1}^{p-1} \psi_j \Delta Y_{t-j} + U_t$$

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Therefore under $H_0 : \varphi = 0$ we simulate from

$$\Delta Y_t = X_t' \beta^* + \sum_{j=1}^{p-1} \psi_j \Delta Y_{t-j} + U_t$$

- p and $\psi_1, \dots, \psi_{p-1}$: “asymptotically negligible”.
Therefore set $p = 1$ ($\psi_j = 0$)

Notes on the ADF simulation process

Therefore under $H_0 : \varphi = 0$ we simulate from

$$\Delta Y_t = X_t' \beta^* + U_t$$

- Distribution of U_t : “asymptotically negligible”.

Notes on the ADF simulation process

Therefore under $H_0 : \varphi = 0$ we simulate from

$$\Delta Y_t = X_t' \beta^* + U_t$$

- Distribution of U_t : “asymptotically negligible”.
We set $U_t \sim N(0, 1)$.

Notes on the ADF simulation process

Therefore under $H_0 : \varphi = 0$ we simulate from

$$\Delta Y_t = X_t' \beta^* + U_t, \quad U_t \sim N(0, 1)$$

- The ADF test is *invariant* to β^* .
(As long as we regress on X_t).

Notes on the ADF simulation process

Therefore under $H_0 : \varphi = 0$ we simulate from

$$\Delta Y_t = X_t' \beta^* + U_t, \quad U_t \sim N(0, 1)$$

- The ADF test is *invariant* to β^* .
(As long as we regress on X_t).
Therefore we set $\beta^* = 0$.

Notes on the ADF simulation process

Therefore under $H_0 : \varphi = 0$ we simulate from

$$\Delta Y_t = U_t, \quad U_t \sim N(0, 1)$$

Summary

- Concepts of critical value and p -value.
- Simulation of critical / p -values by repeated generation from a model under H_0 .
- Dependence of ADF critical / p -values on trend specification.